

Abstract

The smart home system orchestrates a harmonious blend of comfort, security, and sustainability. Through its integration of various features, including temperature regulation, efficient plant watering, and real-time security monitoring, it functions as a reliable ally for your home, easing your responsibilities and enhancing your peace of mind. However, the reliance on technology necessitates preparedness for potential reliability concerns and the occasional need for troubleshooting. While initial investment and ongoing maintenance are necessary, the system's benefits outweigh these considerations. Users may experience an adjustment period as they familiarize themselves with the capabilities and intricacies of the technology, but the rewards of enhanced convenience and safety are well worth the transition.

Introduction

Technology is the future, and as we are approaching this future most of the things are becoming automated. Our project accomplishes the same by automating home appliances like lights, fans, doors, door alarms, and water pumps along with systems to alert the user about potential threats like gas, invasion, high or low moisture levels in plants, etc. Interestingly, all of this can be operated wirelessly by the user with just a single tap on the phone even if the user is far away from home. Although this may sound like a deluxe system that only high-end homes should include, it still has a lot of use cases for common household settings as well. This system can be very helpful for elderly people or people with disabilities who may find it difficult to operate the home in a traditional style. The automated door system removes the nuisance of having a key that could be stolen or misplaced, it also gives users the ability to check and take pictures if someone is at the door and also open/close the door for them even if the user is not at home. Automated lights and fans remove the need to switch them on or off frequently and give users the freedom to switch them on or off anytime. The automated irrigation system can be used to water plants based on their moisture levels, it also includes the facility of being notified if the moisture level is very low (plants need water) or if the moisture level is high (overflow of water).



Construction

We started by making a circuit diagram using online software like Tinkercad, circuit.io, etc. We then did a lot of online research as well as took help from the respective faculty to refine and validate our circuit diagram. We then started with the connections for which we individually tested each component for its voltage and current consumption. For this project, we used a lot of sensors like Gas sensor MQ2, PIR sensor HCSR501, Temperature sensor DHT11, LDR, few buzzers and LEDs, a relay to run the water pump, a motor driver L298N for running the fan motors and giving 5v to other components, and two microcontrollers named ESP32 DEVKIT V1 and ESP32 Cam. We then started connecting the components as per our circuit design and also did the coding side-by-side. Further, we used Google Spreadsheet to store the data online and read and write it with ESP32 using Google Apps script. For UI, we used Appsheet which was easy to connect to our spreadsheet for user interactions with the data. Finally, we made a wooden house and fitted the setup one by one into it.

Results

We were highly successful in creating a smart, secure, and reliable home automation prototype model with an average user response time of 7-8 seconds. This means that it takes approximately 7-8 seconds from the moment the user sends a command until it is executed by the ESP32.

Discussion

Our project integrates various mini-systems into a single home automation project. It focuses on creating an automated interface for users using Google Spreadsheet, Google Apps Script, and AppSheet, instead of other commonly used platforms like Blynk. In this project, we thoroughly studied each component used and their respective connections to produce an effective model with less potential for errors.

Conclusions

In conclusion, our home automation project seamlessly integrates various components including lights, fans, irrigation system, door lock, gas sensor, and motion detectors. Through sophisticated technology and meticulous design, we've created a system that not only enhances convenience but also prioritizes safety and efficiency. By automating these essential aspects of home living, we aim to provide users with a comfortable, secure, and environmentally conscious living environment. With real-time monitoring and intuitive controls, our project exemplifies the potential of modern automation to elevate the quality of life in homes while promoting sustainability and peace of mind.

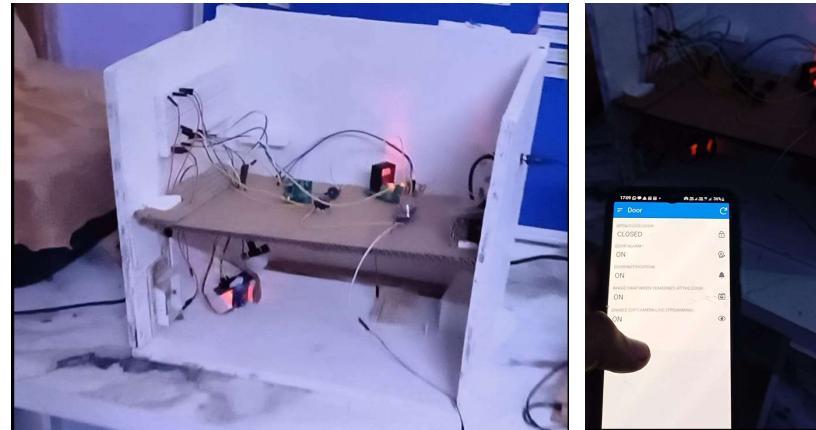


Figure 2. Smart Home Automation System

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